Decision-making Considerations:

- **Financial Viability:** AVK Private Limited should calculate the net present value (NPV) of Project 'M' to determine its financial viability. A positive NPV would indicate that the project is expected to generate value for the company.
- **Risk Assessment:** While the IRR and PI provide valuable insights into Project 'M's potential profitability, the company should also assess the project's risk profile. Factors such as market volatility, technological obsolescence, and regulatory changes could impact the project's performance.

- **Alternative Investments:** AVK Private Limited should evaluate alternative investment opportunities to compare the potential returns and risks associated with Project 'M'. This analysis would help prioritize capital allocation and ensure optimal utilization of resources.

By conducting a comprehensive financial analysis and considering relevant decision-making factors, the company can make an informed decision regarding the feasibility and profitability of undertaking Project 'M'. Continuous monitoring and evaluation of project performance will be essential to ensure alignment with strategic objectives and maximize shareholder value over the project's lifespan.

PV factors at different rates are given below:

Discount factor	15%	14%	13%	12%
1 year	0.869	0.877	0.885	0.893
2 year	0.756	0.769	0.783	0.797
3 year	0.658	0.675	0.693	0.712
4 year	0.572	0.592	0.613	0.636
P.V.A.F for 4 year	2.855	2.913	2.974	3.038

You are required to answer the following questions:

MULTIPLE CHOICE QUESTIONS

1. What is the Cost of Project M?

   (a) ₹ 72.825 Crore

   (b) ₹ 75.943 Crore

   (c) ₹ 71.375 Crore

   (d) ₹ 74.35 Crore

2. What is the present value of cash in flows at cost of capital?

   (a) ₹ 75.943 Crore

   (b) ₹ 100 Crore

   (c) ₹ 26.6 Crore

   (d) ₹ 80.81 Crore

ANSWERS TO MULTIPLE CHOICE QUESTIONS

1. Option (c) ₹ 71.375 Crore

Reason:

Cost of Project

P.V of Cash Out Flow = ₹ 25 Crore x 2.855 = ₹ 71.375 Crore

2. Option (a) ₹ 75.943 Crore

Reason:

PV of Cash inflows

$$PI = \frac{\text{P.V. of CIF}}{\text{P.V. of COF}} = 1.064;$$

$$1.064 = \frac{\text{P.V. of CIF}}{₹\ 71.375};$$

P.V. of CIF = ₹ 75.943 Crore

SOLUTION:

Solution

(i) $P0 = \frac{E(1-b)}{Ke-br}$

$159.09 = \frac{10(1-b)}{0.08-0.12b}$

159.09(0.08 – 0.12b) = 10 – 10b

12.7272 – 10 = 19.0908b – 10b

$b = \frac{2.7272}{0.0908} = 0.30$

Dividend payout ratio = 1- b = 0.70

Dividend per share EPS × (1 – b) = 10 × 0.70 = ₹7

(ii) When r = Ke, There is no optimum ratio.